Deepak Mallampati

APPLIED AI ENGINEER - PYTHON + LLM

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Kent, OH

PROFESSIONAL SUMMARY

Applied AI engineer with 4+ years building production healthcare LLM systems in Python. Ships agentic AI - a conductor-subagent framework (42% less token use) - and the evaluation and safety harness that keeps it reliable (RAGAS, BERTScore, 42% fewer unsafe outputs). Deep healthcare and regulated-domain background: production RAG over clinical corpora and privacy-preserving clinical ML (k-anonymity, l-diversity, differential privacy) with HIPAA discipline. PyTorch/Hugging Face, fine-tuning (LoRA/QLoRA/PEFT), context engineering. US-based, F-1 OPT.

SKILLS

Python + LLM Apps • RAG + Vector Search • LLM Evaluation • Fintech / Regulated • PCI DSS / SOC 2 • PII Masking • Agentic Workflows • FastAPI Services • Model Routing • Audit Logging

Languages: Python, TypeScript, JavaScript, SQL (T-SQL, PostgreSQL, SQLite), Java, C++

AI / ML: PyTorch, scikit-learn, XGBoost, LangChain, LlamaIndex, OpenAI, Anthropic, Hugging Face Transformers, Diffusers, YOLOv8

LLM / GenAI: RAG, agentic workflows, structured outputs, prompt engineering, evaluation pipelines, guardrails, grounding, embeddings, vector search

Backend: FastAPI, Flask, REST, WebSockets, Docker, Celery, Playwright

Frontend: React, Next.js, Vite, Tailwind, Chrome Extensions (Manifest V3)

Data: PostgreSQL + pgvector, MongoDB, Redis, Pandas, NumPy, Snowflake

DevOps: Docker, Jenkins, GitHub Actions, Grafana, Prometheus, AWS (EC2, S3, Lambda)

Domains: Healthcare AI (HIPAA-conscious), Applied AI, Enterprise AI, Workflow Automation, Computer Vision, Multimodal

PROFESSIONAL EXPERIENCE

Progress Solutions Inc. *Jul 2025 - Present*

Jr. AI/ML Engineer Trainee - Agent Engineering, Production LLM

USA - Healthcare Technology, AI Consulting

- Built **Retrieval-Augmented Generation (RAG)** for clinical knowledge retrieval; recursive semantic chunking and transformer embeddings improved retrieval precision **~35%**, reduced irrelevant retrieval **>30%**, response alignment **>90%**.
- Developed **agentic LLM workflows** for multi-step healthcare queries with tool discipline, structured reasoning, and grounding rules; response stability **~25%**, hallucinations cut **>30%**.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show and care engagement scoring; recall **+15-20%** on high-risk categories, precision held **>90%** via class weighting, stratified sampling, threshold calibration.
- Packaged ML/LLM inference as **FastAPI / Flask** services on **Docker** with structured logging and load simulation; post-deploy defects **~30%**.
- EHR preprocessing (Pandas, NumPy); dataset reliability **>98%**, downstream instability **-40%**.
- HIPAA-conscious governance: de-identification, data lineage, evaluation audit trails, stakeholder-facing system-limitation docs.

Energy Solutions International (Emerson) *Jun 2022 - Apr 2023*

Database & DevOps Performance Engineer (Intern)

Hyderabad, India - Oil & Gas ERP

- Optimized **T-SQL stored procedures** on a compliance-sensitive ERP (contracts, nominations, allocations, invoicing); query time **-20%**, retrieval latency **-25%**.
- Built **Jenkins CI/CD pipelines** for schema updates and stored-procedure deployments; deploy errors **-30%**, release cycle **-35 to -40%**.
- Designed performance dashboards with **SQL Server DMVs + Grafana**; incident recurrence **-25%**, root-cause resolution **-30%**.
- Tuned index maintenance and partition-aware queries; reconciliation runtime **-15%**, deadlocks **-15%**.
- Supported RBAC and audit logging for financial modules in a regulated environment.

Suvidha Foundation *Jan 2022 - Mar 2022*

Machine Learning Engineer

Hyderabad, India - Nonprofit, Video Analytics, Applied NLP

- Built **transformer-based video summarization** over 5,000+ recorded sessions; hierarchical summarization and timestamp-aligned clip extraction cut manual review **60-70%**.
- AI-selected highlights aligned with human-curated moments **~85%**.
- Implemented **document Q&A with RAG**; hallucinations **~30%**, contextual accuracy **>85%**.
- Deployed as **Flask** API with a lightweight web interface for non-technical staff.

PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

5-stage multi-agent pipeline (Intake, Validation, Consistency, Duplicate, Risk Scoring) with schema-validated JSON contracts between agents to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policies. Audit-ready explainable risk scoring.

Python, LangChain, FastAPI, pgvector, GPT-4, schema contracts

Manga Lens - Multi-Provider AI Vision Chrome Extension

Real-time manga translation extension shipped to the Chrome Web Store. Multi-provider abstraction (4 AI vision providers: Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama) with per-provider payload handling for worst-case failure isolation. 29 sites supported; 7-day cache; privacy policy and narrowed permissions.

TypeScript, Manifest V3, Service Workers, OpenAI / Anthropic / Ollama

Dream Decoder - Multimodal Creative Platform

End-to-end multimodal pipeline: text description, intermediate prompt-transformation layer, grounded image generation. ~30% contextual alignment gain, ~25-30% first-pass image success.

FastAPI, React + Vite, Tailwind, Perplexity Sonar, Replicate

RESEARCH

Kent State University *Oct 2023 - Jan 2024*

ML & Generative AI Research Assistant

Kent, OH - Privacy-Preserving ML, Healthcare Data

- Built a **privacy-preserving ML pipeline** on clinical hospital-readmission data (k-anonymity k=3, l-diversity l=2, Laplace differential privacy); reduced re-identification risk **3.38% to 0.32%** while retaining **99%** of baseline predictive performance (RF AUC 0.689 vs 0.696).
- Designed a composite **privacy-utility scoring framework** benchmarking six anonymization configurations to identify the optimal operating point.
- Fine-tuned **Qwen3-27B via 4-bit QLoRA** on an NVIDIA H100 for controllable long-form generation; authored an IEEE-format conference paper.

EDUCATION

Master of Science, Computer Science - *Kent State University* *Aug 2023 - May 2025*

Focus: Applied Machine Learning, NLP, Database Systems.

Bachelor of Technology, Computer Science - *KL University* *Jun 2019 - May 2023*

Founder / Lead of E-Farming platform (university project, pilot with 3 cooperatives).

CERTIFICATIONS

Deep Learning Specialization - *DeepLearning.AI (Coursera)* *2024*